15 The graph of the curve C with equation $y = f(x)$ is transformed to give the graph of the curve S with equation $y = f(-x) - 3$

The point on C with coordinates (7, 2) is mapped to the point Q on S.

Find the coordinates of Q.

$y = f(-x)$: reflect in the y-axis.

$(7, 2) \rightarrow (-7, 2)$

1 mark
Reflect x-coordinate.

Then $y = f(-x) - 3$: shift 3 down.

$(-7, 2) \rightarrow (-7, -1)$

1 mark
Then subtract 3 from y.

(-7 , -1)

(Total for Question 15 is 2 marks)

16 Here are the first six terms of a quadratic sequence.

−1 5 15 29 47 69

Find an expression, in terms of n, for the nth term of this sequence.

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(Total for Question 16 is 3 marks)

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